

IP-SAKTI SAHAYAK

Family 8 — Fabaceae / Mulethi (Licorice)

Objective: Create a TKDL-only Fabaceae cluster around Mulethi so IP-SAKTI can identify related TKDL-listed plants and rank them as potential alternatives for further research when a formulation or ingredient triggers a prior-art/IP signal.

Strict TKDL rule: Candidate plants below are restricted to names appearing in the official TKDL plant-name index. The ranking is a prototype similarity score—not proof of medicinal equivalence, patentability, or freedom from infringement.

1. Anchor / main plant

Mulethi / Licorice — Glycyrrhiza glabra Linn.

TKDL Ayurvedic name: Ya■■■imadhu; Unani names include Rubb-al-Soos / Asl-al-Soos; Tamil/common entries include Athimathuram and related names. The TKDL index explicitly lists the plant.

2. Related TKDL species / family candidates — 5

TKDL plant	Relationship	Similarity*	Evidence / model use
Tephrosia purpurea (Linn.) Pers.	Fabaceae; different genus	HIGH — 0.78	TKDL formulation/use evidence; useful cross-genus candidate
Albizia lebbeck (Linn.) Benth.	Fabaceae; different genus	HIGH — 0.78	TKDL examination + formulation evidence
Trigonella foenum-graecum Linn.	Fabaceae; different genus	MEDIUM-HIGH — 0.72	TKDL index + public examination context
Cicer arietinum Linn.	Fabaceae; different genus	MEDIUM — 0.65	TKDL index; specific formulation overlap to verify
Teramnus labialis	Fabaceae; different genus	MEDIUM — 0.65	TKDL index; specific formulation overlap to verify

*Prototype ranking only. Same-family proximity is one feature; specific TKDL use/formulation overlap determines whether a candidate should be surfaced prominently.

3. Official TKDL plant-index verification

Official TKDL plant-name index: <https://www.tkdil.res.in/tkdil/langdefault/common/Utility/Keyword/Plant-Name.asp>

The official TKDL index contains Glycyrrhiza glabra, Tephrosia purpurea, Trigonella foenum-graecum, Cicer arietinum and Teramnus labialis, among many other Fabaceae entries. The public index also lists multiple Albizia and Tephrosia species, allowing this family cluster to be expanded later without leaving TKDL.

4. TKDL evidence — Mulethi / Glycyrrhiza glabra

1197/CHE/2006: TKDL pre-grant opposition brought prior-art references on Glycyrrhiza glabra for osteoarthritis, fracture, stiffness of joints, bone-related disease, dislocation, joint pain, arthritis and arthralgia. The CGPDTM examiner took cognizance of the TKDL references and stated that claims 1–16 were not patentable under Section 3(p) and lacked novelty/inventive step; the application was abandoned.

Official TKDL record: https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/1197_CHE_2006.asp

1111/CHE/2010: TKDL pre-grant opposition included Glycyrrhiza glabra for general tonic, memory enhancement, intellect-promoting and antioxidant/immunopotentiating uses. The record specifically identifies formulations containing licorice as a single/combined ingredient, including a general tonic formulation and a multi-ingredient formulation with Terminalia bellirica, Terminalia chebula, Phyllanthus emblica, Embelia ribes and Piper longum.

Official TKDL record: https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/1111_CHE_2010.asp

EP2110135: TKDL third-party observations cited Glycyrrhiza glabra and Ziziphus jujuba for depression, psychiatric disorders, restlessness and as brain tonic. The EPO examiner record confirms that TKDL prior-art documents were supplied for the application.

Official TKDL record: <https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/EP2110135.asp>

EP2609922: TKDL third-party observations cited formulations containing Glycyrrhiza glabra in combination with Astragalus gummifer for liver diseases including hepatitis, hepatosis, hepatic diarrhoea and hepatic obstruction. The EPO examiner took cognizance of the TKDL references.

Official TKDL record: <https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/EP2609922.asp>

5. Related-candidate evidence

Tephrosia purpurea: In TKDL case 1111/CHE/2010, the plant appears as a single-ingredient formulation for immunopotentiating / anti-ageing / antioxidant use. This makes it a stronger candidate than a plant supported only by the index.

Albizia lebbeck: TKDL case 1642/DEL/2006 concerns a preparation using Albizia lebbeck for headache. TKDL case 1816/CHE/2013 also records an Albizia lebbeck formulation and a claimed composition containing Glycyrrhiza glabra, Albizia lebbeck and other plants for respiratory/common-cold-related conditions. This gives a useful formulation co-occurrence signal.

Official TKDL examination outcome: <https://www.tkdil.res.in/tkdil/Outcomes/1816-CHE-2013.pdf>

Trigonella foenum-graecum: The official TKDL index lists Fenugreek/Methi as a separate Fabaceae entry. A TKDL-linked Patent Office record also includes fenugreek among the ingredients discussed in a food/health formulation context. For IP-SAKTI, keep the candidate at provisional similarity until a formulation/use record specifically relevant to the user's use case is retrieved.

Cicer arietinum: The official TKDL index identifies chickpea/gram as *Cicer arietinum*. It is retained as a verified Fabaceae candidate but should not be promoted solely from family membership.

Teramnus labialis: The official TKDL index identifies *Teramnus labialis* with the Ayurvedic name *Muhapara* and Unani/common terminology. It is retained as a verified family candidate; specific formulation overlap should be retrieved before high-confidence recommendation.

6. Similarity logic for IP-SAKTI

Recommended weighting: botanical relationship + plant part + traditional indication + formulation role + direct TKDL evidence. Same-family candidates begin below same-genus candidates. A candidate with direct TKDL formulation evidence can outrank another candidate with only index presence.

Example: User enters a Mulethi formulation and the system detects relevant prior-art. IP-SAKTI retrieves the Fabaceae cluster, then compares the user's formulation against *Tephrosia purpurea* and *Albizia lebbeck* first because direct TKDL formulation/use evidence is available. It then considers *Trigonella*, *Cicer* and *Teramnus* as lower-confidence candidates until matching evidence is found.

Recommended UI wording: "Potential alternative candidate — same botanical family and supported by TKDL evidence relevant to the selected use. Further botanical, formulation, efficacy and patent/legal verification required."

7. Model-ready dataset fields

family | anchor_plant | candidate_plant | botanical_name | TKDL_name/synonyms | plant_part | traditional_use | formulation | formulation_role | source_text | TKDL_case_id | official_TKDL_url | similarity_score | similarity_reasons | patent_IP_flag | evidence_level | verification_status

8. AI / legal guardrails

- Same family does not mean interchangeable.
- Same family does not establish freedom from patent infringement.
- Never output "no legal barriers" without separate claim-level patent/legal analysis.
- Never infer that a plant is unpatentable simply because TKDL contains traditional-use evidence.
- Separate Section 3(p), novelty and inventive step in the explanation.
- Preserve exact botanical identity and TKDL terminology; do not merge species because vernacular names are similar.
- Show the official TKDL record behind every high-confidence recommendation.

9. Official TKDL source

TKDL official plant-name index: <https://www.tkdil.res.in/tkdil/langdefault/common/Utility/Keyword/Plant-Name.asp>

The public TKDL examiner index also records 1197/CHE/2006 (*Glycyrrhiza glabra* plant extracts), 1642/DEL/2006 (*Albizia lebbeck*), and other relevant examination outcomes. [cite3search3turn3search15](#)